MCU + PROGRAMMING

U1
MSP430G2553IN20
C1
100nF
R12
47k
C2
1nF
J3
PROG / UART
D1
red 3mm
R17
1k

DISPLAY

A1
OLED 128x64 SSD1306
R9
10k
C13
220uF (OLED kit)

MEMORY

U3
23LC1024
C4
100nF
R10
10k
U4
W25Q128 (Adafruit #5634)
C5
100nF
R11
10k

INPUT

U2
SN74HC165N
C3
100nF
R13
1k
SW1
UP
R1
10k
SW2
DOWN
R2
10k
SW3
LEFT
R3
10k
SW4
RIGHT
R4
10k
SW5
SELECT
R5
10k
SW6
START
R6
10k
SW7
B
R7
10k
SW8
A
R8
10k

AUDIO

R14
1k
C6
100nF
C7
10uF
RV1
10k log (PTV09A)
U5
LM386N-1
C10
100nF
C11
10uF
C8
10uF
C9
220uF
C12
47nF (opt)
R16
10R (opt)
R15
10R
J2
Speaker (solder wires)

POWER

J1
LiPo (JST-PH)
A2
USB-C LiPo charger
SW9
EG1218 power
A3
5V boost
A4
3.3V buck-boost
R18
470k
R19
470k
C14
100nF

PWR_FLAG

Generated by scripts/gen_schematic.py from pcb/handheld_design.py — do not hand-edit.
Pin map follows the firmware course: P2.3 SR_LOAD, P2.4 AUDIO_PWM (TA1.2), P2.5 SRAM_CS, P2.6 FLASH_CS.
J3 carries Spy-Bi-Wire (RST, TEST) and the UART to a LaunchPad eZ-FET; same signal order as its J101 block.
R13 isolates the 74HC165's always-driven QH from the shared MISO line. R9–R11 hold the chip selects high during reset.
SW9 grounds PWR_EN (PowerBoost EN + Pololu SHDN) to switch off; it carries no load current.

SPI: P1.5 SCK, P1.6 MISO, P1.7 MOSI. CS: P2.0 OLED, P2.5 SRAM, P2.6 FLASH
MSP430G2553IN20 on board; LaunchPad eZ-FET programs it over Spy-Bi-Wire (J3)

Sheet: /	
File: msp430_handheld.kicad_sch	
Title: MSP430G2553 Handheld — carrier board	
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